

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271890

Kind Code

A1

Publication Date

August 28, 2025

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COMPLEMENTARY TO ABSOLUTE TEMPERATURE (CTAT) CIRCUIT

Abstract

In accordance with an embodiment, a circuit includes: a switched-capacitor voltage divider; a voltage-to-current converter coupled to an output of the switched-capacitor voltage divider, wherein a first output node of the voltage-to-current converter is configured to provide a CTAT current with respect to a reference resistance; a current mirror having an input coupled to a second output node of the voltage-to-current converter, and a diode junction coupled to an output of the current mirror and to an input of the switched-capacitor voltage divider.

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Family ID: 1000008489889

Appl. No.: 19/059716

Filed: February 21, 2025

Foreign Application Priority Data

DE 102024105065.9

Feb. 22, 2024

Publication Classification

Int. Cl.: G05F3/26 (20060101); G01K7/01 (20060101)

U.S. Cl.:

CPC G05F3/267 (20130101); G01K7/01 (20130101);

Background/Summary

[0001] This application claims the benefit of German Patent Application No. 102024105065.9, filed on Feb. 22, 2024, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] This application relates to the field of bandgap voltage reference circuits and, in particular, to a circuit for generating a current that is complementary to absolute temperature (CTAT).

BACKGROUND

[0003] Bandgap voltage reference circuits are circuits that are widely used in voltage regulators and other integrated circuits. They generate an essentially temperature-independent voltage that corresponds to the theoretical bandgap of a semiconductor. With this, it is possible to increase the performance of a circuit in a given temperature range.

[0004] Bandgap reference circuits are based on the current sum of a current $I_{\text{sub.PTAT}}$ that is Proportional To Absolute Temperature (PTAT current) and of a current $I_{\text{sub.CTAT}}$ that is Complementary To Absolute Temperature (CTAT current). Such PTAT and CTAT currents are generated by applying PTAT and CTAT voltages to reference resistors. Summing the two currents makes it possible to cancel out the temperature-dependent terms of the currents and, thus, to obtain a voltage that is essentially independent of the temperature in the desired temperature range.

[0005] For low-power bandgap operations, such as idle (parking) mode operations of a vehicle, very low CTAT currents are desired. However, this requires the reference resistor to have a large resistance and, thus, to occupy a large area.

SUMMARY

[0006] Some embodiments include a switched capacitor voltage divider that is arranged between a diode and the resistor of a CTAT circuit.

[0007] In one example, the disclosure is directed to a circuit comprising a resistor and a diode. The diode is configured to receive a diode current so that a forward voltage occurs across the diode. The circuit further comprises a switched capacitor voltage divider and a voltage-to-current converter. The switched capacitor voltage divider is connected to the diode and configured to receive the forward voltage and to output a scaled voltage that is a fraction of the forward voltage. The voltage-to-current converter is coupled between the switched capacitor voltage divider and the resistor and that is configured to provide a resistor current, which is proportional to the scaled voltage and inversely proportional to a resistance of the resistor. The circuit also comprises a current source configured to provide the diode current dependent on the resistor current.

[0008] In one example, the disclosure is directed to a method comprising the steps of: supplying a diode current to a diode so that a forward voltage occurs across the diode; generating, by a switched capacitor voltage divider, a scaled voltage from the forward voltage; and providing a resistor current, which is proportional to the scaled voltage and inversely proportional to a resistance of a resistor, wherein the diode current depends on the resistor current.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The embodiments described herein can be better understood with reference to the following description and drawings. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the embodiments. Furthermore, in the figures, like reference numerals designate corresponding parts. In the drawings:

[0010] FIG. 1 illustrates one example of a bandgap reference circuit;

[0011] FIG. 2 illustrates one conventional example of a circuit for generating a current that is

complementary to absolute temperature (CTAT circuit);

[0012] FIG. **3** illustrates one example of a circuit structure for a CTAT circuit in accordance with one or more techniques described herein;

[0013] FIG. **4** illustrates another example of circuit structure for a CTAT circuit in accordance with one or more techniques described herein;

[0014] FIG. **5** illustrates a first example of a CTAT circuit in more detail;

[0015] FIG. **6** illustrates a second example of a CTAT circuit;

[0016] FIG. **7** illustrates a third example of a CTAT circuit;

[0017] FIG. **8** illustrates a fourth example of a CTAT circuit which is a modified version of the example of FIG. **7**;

[0018] FIG. **9** illustrates one example of a bandgap reference generator in accordance with one or more techniques described herein, which uses the CTAT circuit of any one of FIGS. **5** to **8**; and

[0019] FIG. **10** is a flowchart illustrating an example method for operating a CTAT circuit, in accordance with one or more techniques described in this disclosure.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0020] Some embodiments provide a circuit for generating a CTAT current for a low-power bandgap reference that advantageously reduces current level and/or resistor area.

[0021] FIG. **1** shows a general example of a bandgap reference voltage circuit **100** configured to generate a reference voltage $V_{sub.BG}$ that is essentially temperature-independent in a desired temperature range. The circuit comprises a first circuit that is configured to generate a current $I_{sub.PTAT}$ that is proportional to absolute temperature (PTAT circuit) and a second circuit that is configured to generate a current $I_{sub.CTAT}$ that is complementary to absolute temperature (CTAT circuit). In order to obtain the currents $I_{sub.PTAT}$ and $I_{sub.CTAT}$, PTAT and CTAT voltages are applied to corresponding reference resistors.

[0022] The circuit further comprises a first summing element that is configured to sum the PTAT current $I_{sub.PTAT}$ and the CTAT current $I_{sub.CTAT}$ to obtain a reference current that is applied to a resistor R . The voltage drop V_R across the resistor R is received as an input by an amplifier that is controlled by a clock signal CK . The amplifier provides the desired bandgap voltage $V_{sub.BG}$ as an output. The circuit further may comprise a second summing element that sums the currents $I_{sub.PTAT}$ and $I_{sub.CTAT}$ and provides a current output.

[0023] FIG. **2** shows an example of a conventional CTAT circuit **200** for generating a CTAT current $I_{sub.CTAT}$. The circuit comprises a resistor **30** having a resistance $R_{sub.CTAT}$ and a diode **10** through which a current $I_{sub.C}$ flows so that a forward voltage $V_{sub.BE}$ occurs across the diode **10**. The forward voltage $V_{sub.BE}$ is referred to as CTAT voltage. The diode **10** may be the base-emitter diode of a bipolar transistor and the forward voltage $V_{sub.BE}$ may thus correspond to the base-emitter voltage of the bipolar transistor. The circuit also comprises a current source that couples the diode **10** and the resistor **30**. In the present example, the current source comprises a current mirror **70** that is configured to supply the resistor **30** with a resistor current $I_{sub.CTAT}$ and also to provide the current $I_{sub.C}$ that flows through the diode **10** dependent on the resistor current $I_{sub.CTAT}$. A voltage drop across the resistor $R_{sub.CTAT}$ is substantially equal to the forward voltage $V_{sub.BE}$. The resistor current $I_{sub.CTAT}$ is thus proportional to the forward voltage $V_{sub.BE}$ and inversely proportional to the resistance $R_{sub.CTAT}$:

$I_{sub.CTAT} = V_{sub.BE} / R_{sub.CTAT}$. The transistors may be configured in such a way that the current $I_{sub.C}$ is equal to the current $I_{sub.CTAT}$.

[0024] Because it is based on a PN junction and uses an actual resistor, the depicted CTAT circuit is particularly robust. Further, it is very easy to implement and does not require any precise clock.

[0025] However, the main drawback of this circuit is its current consumption. The CTAT voltage, which is typically the base-emitter voltage $V_{sub.BE}$ of a bipolar transistor, conventionally ranges from 400 mV to 700 mV over PVT (Process, Voltage, Temperature). This leads to resistor currents $I_{sub.CTAT}$ that may be up to $700 \text{ mV} / R_{sub.CTAT}$.

[0026] For low-power bandgap operations, as needed, for example, during idle mode (parking) of a vehicle, the CTAT current has to be very low. However, in order to obtain CTAT currents that are lower than $1\ \mu\text{A}$, the resistance of the resistor **30** has to be relatively high, e.g. higher than $1\ \text{M}\Omega$. This means that the resistor **30** occupies a large chip area. According to one example, in order to obtain a CTAT current of $500\ \text{nA}$, the resistance of the resistor $R_{\text{sub.CTAT}}$ has to be equal to $1.2\ \text{M}\Omega$.

[0027] The conventional CTAT circuit is thus based on a trade-off between the $R_{\text{sub.CTAT}}$ value, which should be as low as possible, and the CTAT current, which also has to be as low as possible.

[0028] Accordingly, the embodiments described herein aim at solving this trade-off problem, namely to provide a circuit that is able to reduce the CTAT current and the chip area at the same time. In particular, there is a need for a circuit that is robust in the desired operation conditions, namely a supply voltage from 3.3V down to 2V and a temperature range between -40°C . and 175°C .

[0029] There are known solutions, in which a MOS transistor operating in Weak Inversion mode is used to provide the CTAT voltage or in which a MOS transistor operating in linear region is used as the resistor. However these solutions are not reliable in the above-mentioned operation conditions.

[0030] FIG. **3** illustrates an example of a circuit **300** that can be used in a CTAT circuit. The circuit comprises a diode **10** and a resistor **30** that has a resistance $R_{\text{sub.CTAT}}$. A voltage divider **40** is arranged between the diode **10** and the resistor **30**. The voltage divider **40** is configured to reduce the CTAT voltage $V_{\text{sub.BE}}$ across the diode **10** by a factor k down to $V_{\text{sub.BE}}/k$. Specifically, the voltage divider **40** is connected to the diode **10** and is configured to receive the forward voltage $V_{\text{sub.BE}}$ and to output a scaled voltage $V_{\text{sub.BE}}/k$ that is a fraction of the forward voltage $V_{\text{sub.BE}}$. The circuit **300** further comprises a voltage-to-current converter **50** that is coupled between the voltage divider **40** and the resistor **30**. The voltage-to-current converter **50** is configured to provide a resistor current $I_{\text{sub.CTAT}}$ to the resistor **30** such that the voltage across the resistor **30** equals the scaled voltage $V_{\text{sub.BE}}/k$. The resistor current $I_{\text{sub.CTAT}}$ is proportional to the scaled voltage $V_{\text{sub.BE}}/k$ and inversely proportional to the resistance $R_{\text{sub.CTAT}}$ of the resistor **30**. This circuit thus makes it possible to reduce the current $I_{\text{sub.CTAT}}$ by a factor k with respect the solution without voltage divider. As a consequence, it is also possible to reduce the resistance $R_{\text{sub.CTAT}}$ of the resistor **30**, since this reduction can be (at least partly) compensated by the scaling of the voltage $V_{\text{sub.BE}}$. Reducing the resistance $R_{\text{sub.CTAT}}$ leads to a decrease in the required chip space for the resistor **30** and, thus, to a more compact assembly as well as to a reduced power consumption.

[0031] According to one embodiment, the voltage divider **40** is a switched capacitor voltage divider. A switched capacitor voltage divider is a simple solution for producing a scaled voltage. However, the switching operations in the capacitive divider may result in glitches due to charge injection and clock feedthrough.

[0032] FIG. **4** illustrates a further example of a circuit **400** that can be used in a CTAT circuit. This circuit can be considered an improvement/exemplary implementation of the circuit **300** of FIG. **3**. It comprises a switched capacitor voltage divider **40** as the voltage divider. The circuit **400** further comprises an analog filter **60** that is connected between the switched capacitor voltage divider **40** and the voltage-to-current converter **50**. The analog filter **60** is configured to smooth the artifacts produced by the voltage divider operation to produce the scaled voltage $V_{\text{sub.BE}}/k$ with reduced glitches. This smoothed scaled voltage is then fed to the voltage-to-current converter **50**.

[0033] FIG. **5** illustrates a more detailed example of a CTAT circuit **500** for generating a CTAT current $I_{\text{sub.CTAT}}$. It is one possible implementation of the example of FIG. **4**. In addition to the elements already described with regard to FIG. **4**, it further comprises a current source. The different elements of the circuit will be described in further details below.

[0034] In the depicted example, the voltage divider **40** is realized with a passive charge sharing capacitive array of two capacitors. The switched capacitor voltage divider **40** comprises a first

capacitor C.sub.1 with a first end that can be connected to the diode **10** via a first switch S.sub.1 and a second end that is connected to ground. The voltage divider **40** further comprises a second capacitor C.sub.2 with a first end that can be coupled in parallel with the first capacitor C.sub.1 via a second switch S.sub.2 and a second end that is connected to ground. The second capacitor C.sub.2 can either be connected to the ground (to be discharged) via a third switch S.sub.3 or to an output of the voltage divider **40** via a fourth switch S.sub.4. The first capacitor C.sub.1 has a first capacitance C and the second capacitor C.sub.2 has a second capacitance $(k-1) \cdot C$ which is a factor $(k-1)$ larger than the first capacitance C, k being the division ratio of the switched capacitor voltage divider **40**. The voltage divider **40** also comprises a controller with a clock phase generator, which is not shown in FIG. 5 to keep the illustration simple. The controller is configured to control the switching operations of the switched capacitor voltage divider **40** in a conventional way.

[0035] The switched capacitor voltage divider **40** operates as follows:

[0036] During a first time period (Phase 1 as shown in FIG. 5), the first capacitor C.sub.1 is connected to the diode **10** via the first switch S and disconnected from the second capacitor C.sub.2 using the second switch S.sub.2, so that the forward voltage V.sub.BE is applied to the first capacitor C.sub.1. The first capacitor C.sub.1 is thus charged with a charge $Q_{sub.VBE} = C \cdot V_{sub.BE}$. In the same phase, the third switch S.sub.3 is switched on and the fourth switch S.sub.4 is switched off, so that the second capacitor C.sub.2 is connected to ground with both ends. The second capacitor C.sub.2 is thus fully discharged. The reference number “1” in the figure indicates which switches are switched on during the first phase.

[0037] During a second time period (Phase 2 as shown in FIG. 5), the first capacitor C.sub.1 is disconnected from the diode **10** using the first switch S.sub.1 and is connected to the second capacitor C.sub.2 via the second switch S.sub.2. The two capacitors are thus connected in parallel and the charge $Q_{sub.VBE}$ is shared among them. The voltage across the capacitors is thus $V_{sub.X} = Q_{sub.VBE} / (C + C(k-1)) = Q_{sub.VBE} / (k \cdot C) = V_{sub.BE} / k$. Meanwhile, the third switch S.sub.3 is switched off and the fourth switch S.sub.4 is switched on. The voltage divider thus outputs a scaled voltage that is equal to $V_{sub.BE} / k$. The reference number “2” in the figure indicates which switches are switched on during the second phase.

[0038] This operation uses only passive components and does not require any power consumption. The charge-sharing concept has the advantage to be insensible to clock phase inaccuracy. The accuracy of the operation may, however, be limited by the capacitor matching and by charge injection. The charge injection can be minimized by minimizing the size of the switch devices.

[0039] It is clear that the number of capacitors of the switched capacitor voltage divider **40** is not restricted to two and the skilled person would know how to implement further capacitors in the above concept. However, increasing the number of capacitors also increases the control complexity of the voltage divider.

[0040] In the depicted example, the analog filter **60** is a two-stage passive RC low-pass filter, each stage comprising a resistor R.sub.F and a capacitor C.sub.F, wherein the two RC filter stages are coupled in series. According to one example, the resistors R.sub.F may be implemented using high-resistance NMOS transistors operating in linear region. The analog filter **60** is configured to reduce the glitches of the scaled voltage $V_{sub.BE} / k$ output by the switched capacitor voltage divider **40** and to output a smoothed scaled voltage to the voltage-to-current converter **50**.

[0041] In the depicted example, the voltage-to-current converter **50** comprises an operational amplifier **20** and a transistor M.sub.3. The input of the converter **50** is connected to the output of the analog filter **60** and thus receives the smoothed scaled voltage $V_{sub.BE} / k$. An output of the converter **50** is connected to a first end of the resistor **30**. The operational amplifier **20** and the third transistor M.sub.3 are coupled such that the transistor M.sub.3 provides, as resistor current, a current I.sub.CTAT that is proportional to an input voltage V.sub.BE of the voltage-to-current converter and wherein the current I.sub.CTAT has a level such that the voltage drop I.sub.CTAT · R.sub.CTAT across the resistor **30** equals $V_{sub.BE} / k$. In the depicted example, the resistor current

$I_{\text{sub.CTAT}}$ is the same current as the one that flows through the resistor **30** and is thus equal to: $I_{\text{sub.CTAT}} = V_{\text{sub.BE}} / (k \cdot R_{\text{sub.CTAT}})$. The bandwidth of the operational amplifier **20** may be designed to be very small in order to further reduce any residual switching artifacts in the smoothed scaled voltage $V_{\text{sub.BE}}/k$.

[0042] In the depicted example, the transistor $M_{\text{sub.3}}$ is a NMOS transistor. A first input of the operational amplifier **20** receives the smoothed scaled voltage $V_{\text{sub.BE}}/k$ and a second input of the operational amplifier **20** is connected to the source of the transistor $M_{\text{sub.3}}$, which is also connected to the resistor **30**. The drain of the transistor $M_{\text{sub.3}}$ is connected to the current mirror, and the source of the transistor $M_{\text{sub.3}}$ is connected to the first end of the resistor **30**. An output of the operational amplifier **20** is connected to the gate of the transistor $M_{\text{sub.3}}$.

[0043] The circuit **500** also comprises the current mirror that is configured to provide the diode current $I_{\text{sub.C}}$ dependent on the resistor current $I_{\text{sub.CTAT}}$. In the depicted example, the current source comprises a current mirror with a first transistor $M_{\text{sub.1}}$ coupled to the diode **10** and a second transistor $M_{\text{sub.2}}$ coupled to the resistor $R_{\text{sub.CTAT}}$. The first transistor $M_{\text{sub.1}}$ (current mirror output branch) and the second transistor $M_{\text{sub.2}}$ (current mirror input branch) are configured to provide, as diode current $I_{\text{sub.C}}$, a replica of the resistor current $I_{\text{sub.CTAT}}$. According to one example, the transistors $M_{\text{sub.1}}$ and $M_{\text{sub.2}}$ are PMOS transistors. In the depicted example, the source electrodes of both transistors $M_{\text{sub.1}}$ and $M_{\text{sub.2}}$ are connected to each other and to a supply. Further, the gate electrodes of both transistors $M_{\text{sub.1}}$ and $M_{\text{sub.2}}$ are connected to each other. A drain electrode of the first transistor $M_{\text{sub.1}}$ is coupled to the diode **10** and a drain electrode of the second transistor $M_{\text{sub.2}}$ is coupled to the resistor **30** and to the transistor $M_{\text{sub.3}}$ of the voltage-to-current converter. According to another example, the transistors $M_{\text{sub.1}}$ and $M_{\text{sub.2}}$ are identical and the diode current $I_{\text{sub.C}}$ is equal to the resistor current $I_{\text{sub.CTAT}}$.

[0044] With this circuit, it is possible to reduce both the resistor current $I_{\text{sub.CTAT}}$ and the resistance $R_{\text{sub.CTAT}}$ of the resistor **30**. This reduction happens at the cost of the introduction of two active blocks.

[0045] The first active block is the clock phase generator for the switched capacitor voltage divider **40** (not shown). If the full bandgap circuit already comprises a clock phase generator, it can be used for the CTAT circuit. Otherwise, it will have to be added. However, since the concept of the switched capacitor voltage divider **40** is based on charge sharing and since the circuit is using very low-frequency voltages (or even DC voltages), the clock generator does not need to be precise in terms of frequency accuracy or phase jitter. This means that it is possible to use a very low-power oscillator, such as a current-limited ring oscillator. This oscillator can then also be used for other blocks of the full bandgap voltage reference circuit.

[0046] The second active block is the operational amplifier **20** of the voltage-to-current converter **50**. As mentioned above, the bandwidth of the operational amplifier **20** can be designed to reduce the artefacts in the input scaled voltage $V_{\text{sub.BE}}/k$. It is thus possible to use a very small-bandwidth operational amplifier with a very low power consumption.

[0047] In the example of FIG. 5, only passive elements are used in the switched capacitor voltage divider **40**, and the clock for the switching operations is not required to be precise. Further, glitches that may be introduced by the switching operations of the switched capacitor voltage divider **40** can be smoothed by the analog filter **60** and by the use of an operational amplifier **20** in the voltage-to-current converter **50** that has a small bandwidth.

[0048] With the circuit of FIG. 5, because the downscaled forward voltage $V_{\text{sub.BE}}/k$ is applied to the resistor **30**, it is possible to reduce both, the CTAT current $I_{\text{sub.CTAT}}$ and the resistance value $R_{\text{sub.CTAT}}$ of the resistor **30** compared to the solution without voltage divider. For example, if the scaling factor k is equal to 4, it is possible to reduce the CTAT current $I_{\text{sub.CTAT}}$ by two and to reduce the resistance $R_{\text{sub.CTAT}}$ by two as compared with the circuit of FIG. 2. According to one example, the resistance $R_{\text{sub.CTAT}}$ can be taken to be 600 kOhm, which is half of the value of the

resistance in the example of FIG. 2. The current $I_{\text{sub.CTAT}}$ will then be equal to 250 nA, which is also half of the value of current in the example of FIG. 2.

[0049] In the example of FIG. 5, the current mirror (composed of M.sub.1 and M.sub.2 in the depicted example) is connected to a voltage source (supply voltage VDD) or a current source (supply current $I_{\text{sub.BIAS}}$), while the diode **10** and the resistor **30** are connected to the ground. However, it is clear that this situation can be reversed without having to modify the rest of the circuit or losing any advantages of the circuit.

[0050] The CTAT circuit of FIG. 5 can be embedded in a full bandgap voltage reference circuit as illustrated in FIG. 1. Such a full bandgap voltage reference circuit uses the CTAT current $I_{\text{sub.CTAT}}$ of the CTAT circuit for different bandgap operations. It is thus important to be able to source or sink an output current from or by the CTAT circuit.

[0051] FIG. 6 illustrates a further example of a CTAT current circuit **600** that is able to source an output current $I_{\text{sub.OUTPUT}}$. Compared with the circuit of FIG. 5, the circuit **600** comprises a further transistor M.sub.OUTPUT that is arranged in a current mirror configuration with the second transistor M.sub.2 of the current mirror. The control electrodes of the transistors M.sub.2 and M.sub.OUTPUT are connected to each other and the source electrodes of the transistors M.sub.2 and M.sub.OUTPUT are connected to each other. In the depicted example, the further transistor M.sub.OUTPUT is a PMOS transistor. The further transistor M.sub.OUTPUT is configured to mirror the current $I_{\text{sub.CTAT}}$ flowing through the transistor M.sub.2, so that the output current $I_{\text{sub.OUTPUT}}$, which flows through the further transistor M.sub.OUTPUT, is proportional to the current $I_{\text{sub.CTAT}}$. The output current $I_{\text{sub.OUTPUT}}$ can then be used for further bandgap operations. This solution does not change the operation of the CTAT circuit that provides the current $I_{\text{sub.CTAT}}$ and offers a reliable output current. That is, apart from the additional transistor M.sub.OUTPUT the circuit of FIG. 6 is the same as the example of FIG. 5.

[0052] FIG. 7 illustrates another example of a CTAT current circuit **700** that is able to sink an output current $I_{\text{sub.OUTN}}$. Compared with the circuit of FIG. 5, the voltage-to-current converter **50** further comprises an output transistor M.sub.OUTN that is connected in a current mirror configuration to the other transistor M.sub.3 of the voltage-to-current converter **50**. In the depicted example, both the transistor M.sub.3 and the output transistor M.sub.OUTN are NMOS transistors. Specifically, a gate electrode of the output transistor M.sub.OUTN is connected to a gate electrode of the transistor M.sub.3 of the voltage-to-current converter and a source electrode of the output transistor M.sub.OUTN is connected to the source electrode of the transistor M.sub.3. With this, the transistors M.sub.3 and M.sub.OUTN are controlled in the same manner and have the same gate-source voltage. The output current $I_{\text{sub.OUTN}}$ is thus proportional to the resistor current $I_{\text{sub.CTAT}}$. Both the current $I_{\text{sub.CTAT}}$, which flows through the transistor M.sub.3, and the current $I_{\text{sub.OUTN}}$, which flows through the output transistor M.sub.OUTN, also pass through the resistor **30**, so that the total current $I_{\text{sub.RCTAT}}$ that flows through the resistor **30** is equal to: $I_{\text{sub.RCTAT}} = I_{\text{sub.CTAT}} + I_{\text{sub.OUTN}}$. Since the voltage drop $V_{\text{sub.BE}}/k$ across the resistor **30** is not modified by the addition of the output transistor M.sub.OUTN, the resistance $R_{\text{sub.CTAT}}$ of the resistor **30** is equal to: $R_{\text{sub.CTAT}} = (V_{\text{sub.BE}}/k) / (I_{\text{sub.CTAT}} + I_{\text{sub.OUTN}})$. By adding the output transistor M.sub.OUTN, it is thus possible to use a resistor having a reduced resistance value $R_{\text{sub.CTAT}}$, and thus to reduce the chip area required for the resistor **30**. The output current $I_{\text{sub.OUTN}}$ can then be used for further bandgap applications. Apart from the additional transistor M.sub.OUTN, the circuit of FIG. 7 is the same as the example of FIG. 5. Like in the example of FIG. 5, the resistor current $I_{\text{sub.RCTAT}} = I_{\text{sub.CTAT}} + I_{\text{sub.OUTN}}$ is proportional to the scaled voltage $V_{\text{sub.BE}}/k$ and inversely proportional to the resistance $R_{\text{sub.CTAT}}$ of the resistor **30**. The transistor currents $I_{\text{sub.CTAT}}$ and $I_{\text{sub.OUTN}}$ are also proportional to each other.

[0053] FIG. 8 illustrates a further example of a CTAT current circuit **800**, which is a modification/enhancement of the circuit of FIG. 7. Compared with the circuit of FIG. 7, the circuit **800** does not comprise only one output transistor, but two output transistors M.sub.4 and M.sub.5.

The first output transistor M.sub.4 corresponds to the output transistor M.sub.5. OUTN of FIG. 7 and a first output current I.sub.4 flows through it. The second output transistor M.sub.5 is connected to the first output transistor M.sub.4 in a mirror configuration. Each output current I.sub.4, I.sub.5 is proportional to the resistor current I.sub.CTAT and can be used for different bandgap operations. Both the first output current I.sub.4 and the second output current I.sub.5 flow through the resistor **30**. The total current I.sub.RCTAT flowing through the resistor **30** is thus:

$I_{sub.RCTAT} = I_{sub.CTAT} + I_{sub.4} + I_{sub.5}$. It is thus possible to further reduce the resistance value R.sub.CTAT of the resistor **30**. According to one example, the scaling factor k is equal to 4 and the transistors M.sub.4 and M.sub.5 are chosen such that the resistance value R.sub.CTAT is equal to 240 kOhm and the current I.sub.RCTAT that flows through the resistor **30** is equal to 250 nA. Compared to the example of FIG. 2, the current value can be decreased by 50% and the resistance value can be decreased to 20% of the original value. The addition of the output transistors M.sub.4 and M.sub.5 can lead to a lower resistance value R.sub.CTAT, which requires less chip area, and, thus, to a more compact structure. Like in the previous example, the resistor current $I_{sub.RCTAT} = I_{sub.CTAT} + I_{sub.4} + I_{sub.5}$ is proportional to the scaled voltage $V_{sub.BE}/k$ and inversely proportional to the resistance R.sub.CTAT of the resistor **30**. The transistor currents I.sub.CTAT, I.sub.4, and I.sub.5 are also proportional to each other.

[0054] It is clear that the number of additional output transistors is not limited to two and that the skilled person may determine an optimal number of output transistors that can be added to the circuit of FIG. 5, the output transistors being connected in the same way as in FIGS. 7 and 8.

[0055] FIG. 9 shows an example of a full bandgap reference voltage circuit **900** in which the CTAT circuits of FIGS. 5 to 8 can be used. Compared to the circuit of FIG. 1, the bandgap reference voltage circuit **900** further comprises the above-mentioned limited-current oscillator that provides the clock signal for the switched capacitor voltage divider. Because such a voltage divider is based on the concept of charge sharing, the clock phase generator does not need to be particularly precise so that it is possible to use a limited current oscillator. In addition, the circuit comprises a low-power current generator that is configured to generate a reference current I.sub.BIAS for the oscillator and for the CTAT circuit. With this, the full bandgap reference voltage circuit **900** can have a very low power consumption.

[0056] FIG. 10 is a flowchart illustrating an example method **1000** for operating a circuit for generating a CTAT current. The example process **1000** can be employed to operate devices illustrated in this disclosure, such as the circuits according to FIGS. 5 to 8.

[0057] Process **1000** includes supplying a diode current I.sub.C to a diode **10** so that a forward voltage V.sub.BE occurs across the diode (step **1010**). The current I.sub.C may be supplied by a current source. In one example, the current I.sub.C is supplied by a current mirror that comprises two transistors arranged in a mirror configuration. In another example, the diode **10** is the diode of a bipolar transistor and the forward voltage V.sub.BE is the base-emitter voltage of the bipolar transistor.

[0058] The process **1000** further comprises generating, by a switched capacitor voltage divider **40**, a scaled voltage $V_{sub.BE}/k$ from the forward voltage V.sub.BE, k being the scaling factor of the voltage divider **40** (step **1020**). In one example, the switched capacitor voltage divider **40** is connected to the diode **10** and receives the forward voltage V.sub.BE. The switched capacitor voltage divider **40** is connected to a controller with a clock phase generator, which controls the operation of the voltage divider **40**. During a first time, a first capacitor C.sub.1 of the switched capacitor voltage divider **40** is connected to the diode **10** via a first switch S.sub.1 so as to apply the forward voltage V.sub.BE to the first capacitor C.sub.1. In the same time, a second capacitor C.sub.2 of the switched capacitor voltage divider is disconnected from the first capacitor C.sub.1 using a second switch S.sub.2, and is connected to ground through a third switch S.sub.3. During the first time, the first capacitor C.sub.1 is thus charged with a charge $C \cdot V_{sub.BE}$, C being the capacitance of the first capacitor C.sub.1, and the second capacitor C.sub.2 is discharged.

During a second time, the first capacitor C.sub.1 is disconnected from the diode **10** using the first switch S.sub.1 and the second capacitor C.sub.2 is connected via the second switch S.sub.2 to the first capacitor C.sub.1. The charge is thus shared between the capacitors C.sub.1 and C.sub.2. The second capacitor C.sub.2 has a second capacitance $(k-1)*C$, so that the voltage drop across the capacitors is $V_{sub.BE}/k$. The scaled voltage $V_{sub.BE}/k$ is the output of the switched capacitor voltage divider **40**.

[0059] The process **1000** also comprises providing a resistor CTAT current $I_{sub.CTAT}$, which is proportional to the scaled voltage $V_{sub.BE}/k$ and inversely proportional to a resistance value $R_{sub.CTAT}$ of a resistor **30** (step **1030**). The resistor current $I_{sub.CTAT}$ is provided by a voltage-to-current converter **50** that is connected to the output of the switched capacitor voltage divider **40**. The voltage-to-current converter comprises an operational amplifier **20** and a transistor M.sub.3 through which a current $I_{sub.CTAT}$ flows. The transistor M.sub.3 is connected to the resistor **30** and provides the current $I_{sub.CTAT}$ to the resistor **30**. Since the voltage across the resistor **30** is equal to the scaled voltage $V_{sub.BE}/k$, the current flowing through the resistor **30** is equal to $V_{sub.BE}/(k \cdot R_{sub.CTAT})$. In one example, the resistor current $I_{sub.CTAT}$ is equal to the current $V_{sub.BE}/(k \cdot R_{sub.CTAT})$ that flows through the resistor **30**. In another example, the resistor current $I_{sub.CTAT}$ is proportional to the current that flows through the resistor **30**. The current $I_{sub.CTAT}$, which flows through the transistor M.sub.3 of the voltage-to-current converter **50**, is mirrored by the current mirror to provide the diode current $I_{sub.C}$. The diode current $I_{sub.C}$ thus depends on the current $I_{sub.CTAT}$. In one example, the diode current $I_{sub.C}$ is equal to the current $I_{sub.CTAT}$. Since the current flowing through the resistor **30** is reduced, the power consumption can be decreased. In addition, it is possible to use a resistor **30** having a decreased resistance value so as to obtain a more compact assembly.

[0060] In one example, the artifacts that are caused by the switching operations of the switched capacitor voltage divider **40** are smoothed by a filter **60**, in particular an analog passive filter, arranged between the switched capacitor voltage divider **40** and the voltage-to-current converter **50**. In addition, the operational amplifier **20** of the voltage-to-current converter **50** may be a small-bandwidth operational amplifier that further reduces the artifacts of the scaled voltage $V_{sub.BE}/k$.

[0061] In one example, the current $I_{sub.RCTAT}$ flowing through the resistor **30** is the sum of the current $I_{sub.CTAT}$ flowing through the transistor M and of at least a further current $I_{sub.OUTN}$, $I_{sub.4}$, $I_{sub.5}$ that flows through an output transistor M.sub.OUTN, M.sub.4, M.sub.5 of the voltage-to-current converter **50**. The at least one output transistor M.sub.OUTN, M.sub.4, M.sub.5 is connected to the main transistor M.sub.3 of the voltage-to-current converter **50** in a mirror configuration. The output currents $I_{sub.OUTN}$, $I_{sub.4}$, $I_{sub.5}$ are proportional to the resistor current $I_{sub.CTAT}$ and can be used for further bandgap operations. Further, since the addition of the output current increases the total current flowing through the resistor **30**, it is possible to further decrease the resistance value $R_{sub.CTAT}$ of the resistor **30**.

[0062] The present application describes the use of a switched capacitor voltage divider in a circuit for providing a CTAT current. This circuit is configured to be used in a bandgap voltage reference circuit and usually comprises a diode and a resistor. The current flowing through the resistor is proportional to the CTAT current. According to the application, the switched capacitor voltage divider is connected between the diode and the resistor. Since the switched capacitor voltage divider provides a scaled voltage to the resistor, it is possible to reduce both the CTAT current and the resistance value of the resistor. With this, the current consumption can be kept low and the chip area dedicated to the resistor can be decreased, leading to a more compact assembly. The switched capacitor voltage divider only has any passive elements and does not require a precise clock.

Switching artifacts of the scaled voltage can be smoothed by using a passive analog filter and a voltage-to-current converter having a small-bandwidth operational amplifier. The resulting circuit has a very low power consumption and can be used for low-power bandgap applications.

[0063] Although various embodiments have been illustrated and described with respect to one or

more specific implementations, alterations and/or modifications may be made to the illustrated examples without departing from the spirit and scope of the features and structures recited herein. With particular regard to the various functions performed by the above described components or structures (units, assemblies, devices, circuits, systems, etc.), the terms (including a reference to a “means”) used to describe such components are intended to correspond—unless otherwise indicated—to any component or structure that performs the specified function of the described component (e.g., that is functionally equivalent), even if it is not structurally equivalent to the disclosed structure that performs the function in the herein illustrated exemplary implementations of the present disclosure.

Claims

1. A circuit comprising: a resistor; a diode configured to receive a diode current so that a forward-voltage occurs across the diode; a switched capacitor voltage divider that is connected to the diode and configured to receive the forward-voltage and to output a scaled voltage that is a fraction of the forward-voltage; a voltage-to-current converter that is coupled between the switched capacitor voltage divider and the resistor and that is configured to provide a resistor current proportional to the scaled voltage and inversely proportional to a resistance of the resistor; and a current source configured to provide the diode current dependent on the resistor current.
2. The circuit of claim 1, wherein the diode is base-emitter junction of a bipolar transistor.
3. The circuit of claim 1, wherein the current source comprises: a first transistor coupled to the diode, and a second transistor coupled to the resistor, wherein the first transistor and the second transistor form a current mirror configured to mirror the resistor current to the diode current.
4. The circuit of claim 1, wherein the switched capacitor voltage divider comprises a controller, a first capacitor, a second capacitor configured to be coupled in parallel with the first capacitor, wherein the controller is configured to: during a first time period: connect the first capacitor to the diode via a first switch so as to apply the forward-voltage to the first capacitor, disconnect the second capacitor from the first capacitor using a second switch, and connecting the second capacitor to ground using a third switch, and during a second time period: disconnect the first capacitor from the diode using the first switch, and connect the second capacitor to the first capacitor via the second switch.
5. The circuit of claim 4, wherein the first capacitor comprises a first capacitance and the second capacitor comprises a second capacitance which is a factor $k-1$ larger than the first capacitance, wherein k is a division ratio of the switched capacitor voltage divider.
6. The circuit of claim 1, further comprising an analog filter connected between the switched capacitor voltage divider and the voltage-to-current converter, the analog filter configured to smooth artifacts produced by operations of the switched capacitor voltage divider to produce the scaled voltage.
7. The circuit of claim 1, wherein the voltage-to-current converter comprises an operational amplifier and a third transistor, wherein the operational amplifier and the third transistor are coupled such that the third transistor provides, as the resistor current, a current that is proportional to an input voltage of the voltage-to-current converter.
8. The circuit of claim 7, further comprising at least one output transistor, wherein the at least one output transistor is connected to third transistor of the voltage-to-current converter such that the third transistor and the at least one output transistor form a current mirror configured to provide an output current that is a replica of the resistor current.
9. A full bandgap reference circuit, comprising: the circuit of claim 1, and a further circuit that is configured to output a current that is proportional to an absolute temperature with respect to a reference resistor, wherein the full bandgap reference circuit is configured to sum the resistor current and the current output by the further circuit.

- 10.** A method, comprising: supplying a diode current to a diode so that a forward voltage occurs across the diode; generating, by a switched capacitor voltage divider, a scaled voltage from the forward voltage; and providing a resistor current proportional to the scaled voltage and inversely proportional to a resistance of a resistor, wherein the diode current depends on the resistor current.
- 11.** The method of claim 10, further comprising: during a first time, connecting a first capacitor of the switched capacitor voltage divider to the diode via a first switch to apply the forward voltage to the first capacitor, disconnect a second capacitor of the switched capacitor voltage divider from the first capacitor using a second switch, and connect the second capacitor to ground through a third switch, and during a second time, disconnect the first capacitor from the diode using the first switch and connect the second capacitor to the first capacitor via the second switch.
- 12.** A circuit, comprising: complementary to absolute temperature (CTAT) circuit comprising: a switched-capacitor voltage divider, a voltage-to-current converter coupled to an output of the switched-capacitor voltage divider, wherein a first output node of the voltage-to-current converter is configured to provide a CTAT current with respect to a reference resistance, a current mirror having an input coupled to a second output node of the voltage-to-current converter, and a diode junction coupled to an output of the current mirror and to an input of the switched-capacitor voltage divider.
- 13.** The circuit of claim 12, wherein the CTAT circuit further comprises an analog filter coupled between the output of the switched-capacitor voltage divider and the input to the voltage-to-current converter.
- 14.** The circuit of claim 12, wherein the voltage-to-current converter comprises: a first transistor having an output node coupled to the input of the current mirror; a resistor coupled to a reference node of the first transistor, wherein the resistor comprises the reference resistance; and an amplifier having a first input coupled to the output of the switched-capacitor voltage divider, a second input coupled to the reference node of the transistor, and an output coupled to the reference node of the first transistor.
- 15.** The circuit of claim 14, wherein the voltage-to-current converter further comprises a second transistor having a control node coupled to the output of the amplifier and an output node forming the first output node of the voltage-to-current converter.
- 16.** The circuit of claim 12, further comprising: a proportional to absolute temperature (PTAT) circuit; and a current summing node coupled to a current output of the PTAT circuit and the first output node of the voltage-to-current converter.
- 17.** The circuit of claim 16, further comprising an output resistor coupled to the current summing node.
- 18.** The circuit of claim 17, further comprising an amplifier having an input coupled to the output resistor, wherein the amplifier is configured to output a bandgap reference voltage.
- 19.** The circuit of claim 17, further comprising a limited current oscillator having an clock output coupled to a clock input of the switched-capacitor voltage divider.
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